

INDUSTRIAL ETHERNET NETWORKING:

PLANT AUTOMATION AND SCADA ASPECTS

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To Dr. Jean-Paul Prevost

Enclosed with this letter is a copy of the technical report for industrial ethernet networking. Our team is submitting this report to fulfill the requirements of the ELEC2401 technical project course at NAIT. The topic discussed in the report was approved September 12, 2025

The enclosed report provides the design and implementation of the SCADA network prototype. It includes an overview of essential SCADA concepts, critical networking fundamentals and a detailed description of the network topology for Lab A185. Once those fundamentals are established, the report outlines specifics on the Configuration of field devices including routers, VFD, and PLCs. In addition, it discusses the Software/ drivers required to operate the devices and the key features utilized. Lastly, Included are the key findings and recommendations for improving the specific elements for the SCADA network prototype.

Thank you for reading our report on Industrial Ethernet Networking as well as choosing Energized Electrical to conduct this project. If you would like to discuss the findings in more detail or if there is any additional information needed, kindly reach out to the team through email or phone for a consultation. We look forward to working with you further.

Sincerely,

Yuvraj Grewal

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Enclosure: Technical Report
Cc: Mr. Tegh Manesh
Ms. Julie Schenk

Abstract

This report presents the design, implementation, and evaluation of a SCADA network prototype developed for LAB A185. The project focuses on establishing a functional model of an industrial control environment incorporating PLC, VFD, industrial routers, and associated software. It begins with exploring core SCADA concepts required to understand the projects network topology. It then explains to the fundamental network concepts that are required for router configuration. Field device configuration is also covered to ensure proper integration and communication within the prototype network. The project integrates key software tools such as Ignition SCADA for visualization, Studio 5000 for PLC programming and Rockwell Drive Tools for VFD management. The project also covers essential software extensions including EDS files, AOP, and drivers. The resulting system provides a comprehensive demonstration platform for modern industrial automation, emphasizing the communicational abilities of devices and the capabilities of control design.

TABLE OF CONTENTS

List of Figures	v
List of Tables	vi
Acknowledgements.....	vii
1. Introduction.....	1
1.1 Purpose.....	1
1.2 Scope.....	2
2. SCADA	3
2.1 SCADA Host	4
2.2 SCADA Communication	4
2.3 SCADA Sites	4
3. Network Topology	6
3.1 Routers	7
2.1.1 MGuard 2102 router.....	7
3.1.2 TC Router 3002T-4G-Router.....	8
3.1.3 D-Link AC1750 router.....	9
2.2 Router Configurations.....	9
2.2.1 Network Address Translation (NAT)	9
3.2.2 Static Route.....	11
3.3 Network Security	13
4.0 Field devices	15
3.1 PowerFlex 70 EC-E Variable Frequency Drive (VFD)	15
3.2 Power Flex 753 Variable Frequency Drive (VFD).....	18
4.0 Programing Software Configuration and setup.....	21
4.1 Rockwell Software suites.....	21
4.1.1 Rockwell Network Browsers	21
4.1.2 STUDIO 5000.....	22
4.2.1 Drive Tools	25
4.2.2 AOP's, EDS, And Firmware.....	26
4.2 Ignition.....	27
5.0 Conclusion	30
References.....	31
Appendix A - Definitions.....	34

List of Figures

Figure 1: SCADA Diagram	3
Figure 2: Network Topology for A185	6
Figure 3: MGuard 2102 Router.....	7
Figure 4: TC Router 3002T-4G-Router	8
Figure 5: D-Link AC1750 Router.....	9
Figure 6: NAT Configuration	10
Figure 7: ICMP packets sent through NAT	10
Figure 8: MGurad Web Interface for static route.....	11
Figure 9: ICMP packets sent through Static Route.....	12
Figure 10: Firewall rule in mGuard	13
Figure 11: ICMP packets with the firewall.....	14
Figure 12:Power Flex 70.....	15
Figure 13: 20-COMM-E	16
Figure 14: PowerFlex 753.....	18
Figure 15: 20-750-ENETR	19
Figure 16: Factory Linx Network browser	22
Figure 17: Studio 5000 753 Connection page.....	23
Figure 18 Studio 5000 70 Connection page.....	24
Figure 19: Drive Executive.....	25
Figure 20: Rockwell Automation Files For VFD	26
Figure 21: Ignition Adding Devices.....	28
Figure 22: Ignition Device Connected.....	28
Figure 23: Ignition Perspective	29
Figure 24: Tag Brower.....	29

List of Tables

Table 1: 20-COMM-E settings 16

Table 2: Power Flex 70 Parameters 17

Table 3: Power Flex 753 Parameters 19

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Thanks to their guidance and industry knowledge we were able push this project to completion. We are thankful to be given the opportunity to work on this project and expand our knowledge in this cutting-edge technology.

1. INTRODUCTION

Industrial technology is changing from serial to Ethernet, and newer ways to communicate between devices are being implemented to better enhance industrial processes. Ethernet provides faster and more reliable connection with protocols like TCP/IP (Transmission Control Protocol/Internet Protocol) than older serial communication. Therefore, companies such as Allen Bradley are inventing newer technologies like EtherNet/IP to communicate between devices. Furthermore, with the addition of SCADA, it has become easier to monitor the entire plant processes in real time and have supervisory control over the whole process.

SCADA stands for Supervisory Control and Data Acquisition; it is a control system that is configured to monitor and store data from the PLCs and to provide unified control over the entire process. The report discusses the SCADA simulation conducted in the A185 lab using Ignition software to run a SCADA host. Moreover, the Graphical User Interface (GUI) is built using Ignition for real-time data monitoring and easier user control.

The lab simulation also demonstrates a network architecture of a real industrial plant environment. It highlights how industrial networking devices are used to connect multiple networks, manage data flow between them, and maintain secure communication. In addition, the report also looks into the automation aspects of the plant network, with an emphasis on monitoring and controlling field devices. It outlines how data is collected from industrial equipment and made available to SCADA systems for supervision and control. Overall, the report illustrates how networking, automation, and SCADA systems work together to support efficient and integrated industrial operations.

1.1 Purpose

The purpose of the project is to create a SCADA model that allows for Electrical Engineering Technology (ELT) students at Northern Alberta Institute of Technology (NAIT) to receive education on the basic idea of SCADA systems by simulating a real-world scenario of a large plant by using the equipment available in NAIT's lab. This report talks about a full-scale model of a SCADA system that was made in lab using basic principles of SCADA to match what is used within the industry. With objective to configure routers to be able to send the power, current, and voltage that is being consumed by the motors across the networks into one location. This is to

simulate a real plant, routing between networks so that the hosts from one network can talk across to other networks as is done in industrial automation.

1.2 Scope

This report will cover communication methods covered include Network Address Translation (NAT), adding Static Routes, and firewall policies implemented into routers. The programming of the PLCs, connecting the PLCs to the VFDs and sending data from one PLC to another using Studio 5000 is also covered. Moreover, it will look at the various tools and applications such as Ignition software that is used to make and build a Graphical User Interface (GUI) to simulate an industrial network.

This report will focus specifically on building the SCADA network using the following:

- Allen Bradley Programmable Logix Controller (PLC)
- Power Flex 70 Variable Frequency Drive (VFD)
- Power Flex 735 Variable Frequency Drive (VFD)
- Cisco Catalyst 3950 switch
- Phoenix contact mGuard 2102 router
- D-Link AC1750 router

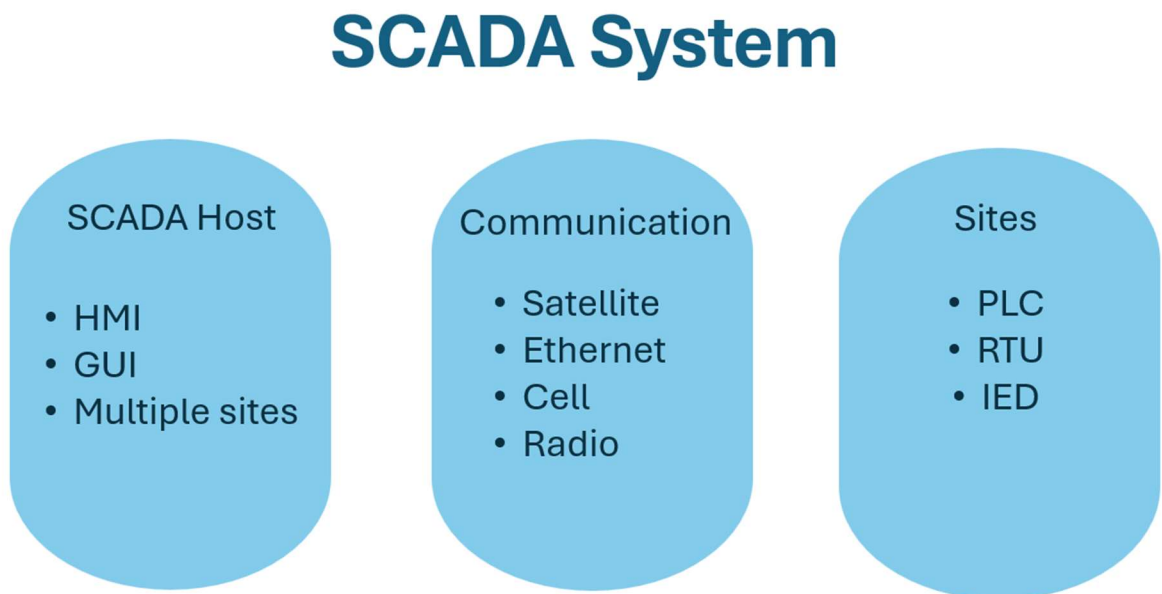
This report only focuses on communicating the PLC to two different VFD and does not include their specific features and abilities. The VFDs are only used for acquiring real time data from the field and forwarding it to the PLC. The report does not include adding OPC servers for data acquisition over the network and making VPN tunnel between networks.

2. SCADA

SCADA stands for supervisory control and data acquisition. It is a control system that uses software and hardware to control processes that are spread across a large geographical area. SCADA systems are the foundation of the automation industry allowing processes to operate, monitoring and controlling machinery for peak efficiency. To further understand the components of the SCADA system that can divide SCADA into 3 main components:

- SCADA Host
- Communication
- Sites

Figure 1: SCADA Diagram



Note. Sourced from Mark Priess (2025, October). NAIT INSTRUMENTATION
ENGINEERING TECHNOLOGY

2.1 SCADA Host

The SCADA host is the point which houses central control of the system. This is typically a computer with software to monitor, control, and store data. This data can be stored onto a database or local server through the host. Ignition is a server-centric solution-centric service developed for SCADA systems (Inductive Automation n.d.) and is the project's example of a SCADA host.

2.2 SCADA Communication

Communication is the way to connect to remote sites and is the focus of this project. This is the network that spans across multiple sites where communication is needed. There are different forms of communication such as cellular, radio, ethernet, and satellite that are used within industrial networking. Things like a cell phone and home internet are examples of the networks you use every day. The project visualizes and investigates the different devices used on its network to get a better understanding of network communication.

2.3 SCADA Sites

Lastly are the sites in which the equipment running the processes is operating and where the data is collected via the field devices. This can include programmable logic controllers (PLC's), variable frequency drives (VFD's), and remote terminal units (RTU's), sensors, motors and any devices a process requires for operation.

In the project model, the use of VFDs and PLCs will simulate the plant in which these essential processes operate. The VFD is used to capture and control the essential data from motors running in large scale plants. PLCs are used to program control logic to operate multiple VFD's. This acts like the central point in which the data is stored from the site. This data is captured into the PLC as controller tags that allow for the design of HMI on the SCADA host side.

SCADA is done in large plants for three main reasons:

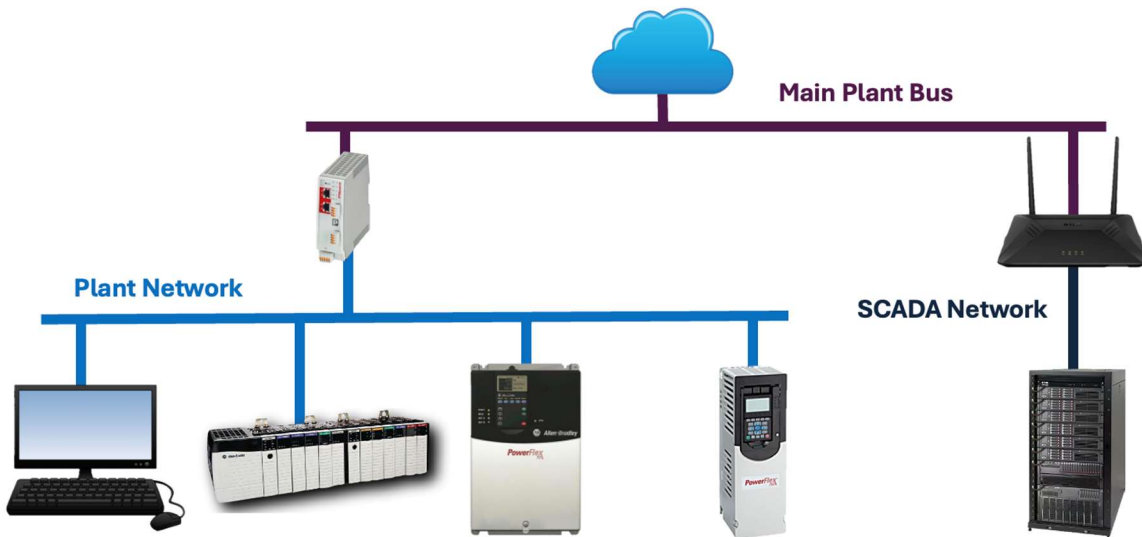
1. Manage traffic: In a large plant there can be countless devices that are constantly sending or receiving data and by putting them into different networks it can help manage the traffic more efficiently.
2. Managing access: Plants can have hundreds of employees working at the same time and giving them all access to the whole network can pose a significant risk.
3. Supervision: When you have many processes going on at the same time it becomes important to put all the data in one location to easily supervise the whole plant.

The biggest example of SCADA within Alberta is the Oil & Gas industry, since Alberta has Oil reserves across the province including Athabasca, Cold Lake and Peace River (Oil Sands Facts and Statistics, 2025). Therefore, it is crucial to be able to monitor how much oil is being produced and transported across other provinces.

3. NETWORK TOPOLOGY

This section discusses the configuration of the routers to securely forward data packets between multiple networks. **Figure 2** shows the full-scale network model, along with all the different components. The network is divided into three sections, each with different network addresses.

Figure 2: Network Topology for A185



The Plant Network, which accommodates the field equipment such as PLCs, and VFDs that collect data. This network is given the network designation of 192.168.1.0/24 with mGuard as their default gateway. The SCADA network which is the 10.10.10.0/24 network, this has the SCADA host with the default gateway as D-Link AC1750 and communicates to the Plant Network through the main bus.

3.1 Routers

Initially there were three routers chosen for this simulation, but due to the limited capabilities of the TC Router it was later excluded out of the network. The list of the routers along with their model number are:

- Phoenix Contact mGuard 2102
- Phoenix Contact TC Router 3002T-4G-Router
- D-Link AC1750

2.1.1 MGuard 2102 router

MGuard router is a part of Phoenix Contact's security series router that is made of rough industrial environments, it is IP20 rated with temperature range of -20°C to 60°C. The router is used in SCADA system because of its VPN and remote maintenance features along with secure firewall and its redundant power source (Phoenix Contact, 2025b).

Figure 3: MGuard 2102 Router



In the lab simulation it was used to do 1:1 NAT between the control network and factory floor network. It is also used to make a static router between the DP-Link router to communicate to the SCADA network. Lastly, it also has firewall rules to restrict unwanted access along with log of packets sent through the router.

3.1.2 TC Router 3002T-4G-Router

TC Router is a 4G cellular router that uses cellular antenna to have high speed internet, up to 150Mbps instead of fiber optic. It is made for harsh temperatures between -40°C to +70°C and has many features that make it feasible for a remote SCADA system. These features include IPsec and OpenVPN, VPN remote start, logbook, and alert SMS etc.

Figure 4: TC Router 3002T-4G-Router



This router was initially planned to use for IPsec and OpenVPN used for VPN tunneling to connect from the router to the mGuard to access the factory floor network through the internet. Later it was realized that the router used a special data only SIM with a public IP that is only obtainable through purchase with a carrier. In addition to that, the mGuard also required a public IP address provided by the ISP so that both can tunnel to each other through the internet. Therefore, the idea was later changed, and a static route was configured between the factory floor network and SCADA network.

3.1.3 D-Link AC1750 router

D-Link router is a normal home router which is both a gateway and a wireless access point. It acts as the default gateway for the SCADA host and simply forwards all the data packets from the SCADA network and the mGuard router. The D-Link AC1750 router is a residential router does not have advanced routing capabilities like Phoenix Contact routers.

Figure 5: D-Link AC1750 Router



2.2 Router Configurations

All devices in the network uses TCP/IP; it stands for Transmission Control Protocol/Internet Protocol. TCP/IP is a set of communication protocols that enables devices like computers, PLC and VFD to communicate over a network (Fortinet, n.d.).

2.2.1 Network Address Translation (NAT)

1:1 NAT maps the IP addresses of a real network to virtual IP addresses, devices in the real network can therefore be accessed directly via their assigned (mapped) virtual IP addresses on the WAN side (mGuard Configuration Examples, 2018).

In **Figure 6** the router setting of the mGuard 2102 router is shown is shown for 1:1 NAT. This illustrates that the IP address of the PLC (192.168.1.124) on the Plant Network is being changed to a virtual IP address of 172.16.50.250 on the WAN side.

Figure 6: NAT Configuration

Seq	Real network	Virtual network	Netmask	Enable ARP	Comment
1	192.168.1.124	172.16.50.250	32	☒	

Figure 7 showcases the ping test conducted to verify a stable connection between the device on the WAN side and the PLC on the LAN side using NAT. This confirms that the configuration was completed correctly and that the router not only created the virtual IP, but is also forwarding packets from the virtual IP to the real IP address. NAT is useful as it does not require any changes to the company network by adding routes through multiple routers. NAT also hides the real IP address of the host from the outside network or allowing only selected equipment visible to the outside network.

Figure 7: ICMP packets sent through NAT

```
C:\WINDOWS\system32\cmd. X + v
C:\Users\ygrewall>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : nait.ca
    Link-local IPv6 Address . . . . . : fe80::38dd:dd99:6bca:347f%15
    IPv4 Address. . . . . : 10.131.12.244
    Subnet Mask . . . . . : 255.255.254.0
    Default Gateway . . . . . : 10.131.13.254

Ethernet adapter PLC:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::2570:8adb:743d:8e3c%11
    IPv4 Address. . . . . : 172.16.50.76
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

C:\Users\ygrewall>ping 172.16.50.250

Pinging 172.16.50.250 with 32 bytes of data:
Reply from 172.16.50.250: bytes=32 time=1ms TTL=63
Reply from 172.16.50.250: bytes=32 time=1ms TTL=63
Reply from 172.16.50.250: bytes=32 time=1ms TTL=63
Reply from 172.16.50.250: bytes=32 time=1ms TTL=63

Ping statistics for 172.16.50.250:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    approximately 0.000ms round trip time for each of the 4 replies
```

3.2.2 Static Route

For communication between the SCADA network on the factory floor, an internal route was added between the two gateways. This essentially means programming the routers to know where to forward the data packets for another network. This allows communication between devices over multiple networks using the real IP address.

Unlike NAT—where traffic typically flows from the WAN toward the LAN using virtual IP, static routing allows bidirectional communication between both networks.

Figure 8 is the Web Interface of the mGuard 2102 router add shows the configuration done to add the static route using the mGuard.

Figure 8: MGurad Web Interface for static route

The screenshot displays the mGuard 2102 router's web interface. On the left is a navigation menu with categories: Management, Network, Interfaces, Authentication, Network Security, IPsec VPN, OpenVPN Client, Logging, and Support. The 'Network' section is expanded, showing 'Interfaces'. The 'Internal' tab is selected. The 'Internal Networks' table has columns for Seq., IP address, Netmask, Use VLAN, and VLAN ID. A single entry is shown with Seq. 1, IP address 192.168.1.1, Netmask 255.255.255.0, Use VLAN unchecked, and VLAN ID 1. Below this is the 'Additional Internal Routes' section with columns for Seq., Network, and Gateway. A single entry is shown with Seq. 1, Network 10.10.10.0/24, and Gateway 10.10.10.254.

Seq.	IP address	Netmask	Use VLAN	VLAN ID
1	192.168.1.1	255.255.255.0	☐	1

Seq.	Network	Gateway
1	10.10.10.0/24	10.10.10.254

Figure 8 states at any packet request for a 10.10.10.0/24 network shall be passed to the gateway 10.10.10.254, which is the D-Link. The mGard and the D-Link are connected to the main bus via a ethernet cable, packets flow from the mGuard to the D-link through the pain bus.

The configuration was tested by conducting a ping test from the SCADA host to the PLC, both being in their respective networks. **Figure 9** shows the ICMP packets sent from MacBooks terminal in the SCADA network, along with its IP address and the route taken by the packet.

Figure 9: ICMP packets sent through Static Route

```
yuvrajsinghgrewal@Yuvrajs-MacBook-Air-49 ~ % ifconfig en7
en7: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=6464<VLAN_MTU,TSO4,TSO6,CHANNEL_IO,PARTIAL_CSUM,ZEROINVERT_CSUM>
    ether 4c:ea:41:68:8e:1c
    inet6 fe80::1440:94dc:2519:5288%en7 prefixlen 64 secured scopeid 0x14
    inet 10.10.10.10 netmask 0xfffff00 broadcast 10.10.10.255
    nd6 options=201<PERFORMNUD,DAD>
    media: autoselect (100baseTX <full-duplex>)
    status: active
yuvrajsinghgrewal@Yuvrajs-MacBook-Air-49 ~ % ping 192.168.1.124
PING 192.168.1.124 (192.168.1.124): 56 data bytes
64 bytes from 192.168.1.124: icmp_seq=0 ttl=62 time=1.676 ms
64 bytes from 192.168.1.124: icmp_seq=1 ttl=62 time=1.933 ms
64 bytes from 192.168.1.124: icmp_seq=2 ttl=62 time=1.548 ms
64 bytes from 192.168.1.124: icmp_seq=3 ttl=62 time=1.837 ms
64 bytes from 192.168.1.124: icmp_seq=4 ttl=62 time=1.688 ms
^C
--- 192.168.1.124 ping statistics ---
5 packets transmitted, 5 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 1.548/1.720/1.933/0.144 ms
yuvrajsinghgrewal@Yuvrajs-MacBook-Air-49 ~ % traceroute 192.168.1.124
traceroute to 192.168.1.124 (192.168.1.124), 64 hops max, 40 byte packets
 1  10.10.10.254 (10.10.10.254)  1.115 ms  0.692 ms  0.515 ms
 2  172.16.50.200 (172.16.50.200)  1.488 ms  0.944 ms  0.843 ms
 3  192.168.1.124 (192.168.1.124)  1.196 ms  1.029 ms  1.038 ms
yuvrajsinghgrewal@Yuvrajs-MacBook-Air-49 ~ %
```

The last command “traceroute” in **Figure 9** shows the router taken by the ICMP packet:

- In every case the packets are first sent to the Local Network, to see if the host is within the network.
- The packet is sent to the default gateway, which is the D-Link router with an IP address of 10.10.10.254
- The router looks at its router table and sends the packets to mGuard with IP of 172.16.50.200
- Finally, the mGuard forwards the packets to the destination, the PLC (192.168.1.124)

The process will be the opposite if packets were sent from the factory floor network to the SCADA network.

3.3 Network Security

The plant router most commonly uses a reject all firewall policy to block all traffic from the WAN to talk to the PLC. The reason for this is that PLC do not have any built-in authentication and anyone who can access the PLC can change the program. Therefore, blocking all traffic and then only allowing selected IP address to pass gives tighter control.

Firewalls rules were added in mGuard to restrict access, so that only allowed persons/users can program the PLC as shown in **Figure 10**.

Figure 10: Firewall rule in mGuard

The screenshot displays the mGuard configuration interface for Network Security > Packet Filter. The left sidebar contains a navigation menu with options: Management, Network, Interfaces, Ethernet, NAT, DNS, DHCP, Proxy Settings, Dynamic Routing, Authentication, Network Security, Packet Filter, IPsec VPN, OpenVPN Client, Logging, Settings, Browse Local Logs, and Support. The main content area is titled 'Incoming Rules' and includes tabs for 'Incoming Rules', 'Outgoing Rules', and 'IP/Port Groups'. Under the 'Incoming Rules' tab, there is a 'General firewall setting' section with a dropdown menu set to 'Use the firewall ruleset below'. Below this is a table with columns: Interface, Protocol, From IP, From port, To IP, To port, and Action. The table contains one rule: Interface: All, Protocol: All, From IP: 172.16.50.76/32, To IP: 192.168.1.124/32, and Action: Reject. At the bottom, there is a checkbox for 'Log entries for unknown connection attempts' which is checked.

Interface	Protocol	From IP	From port	To IP	To port	Action
All	All	172.16.50.76/32		192.168.1.124/32		Reject

The Firewalls is set to block all incoming packets from the 172.16.50.76 IP address that are being sent to 192.168.1.124 (PLC) address. This rule can be changed to block an entire network by changing the subnet mask, the /32 at the end specifies the subnet and can be changed as required.

To test the firewall ICMP packets were done as show as in Figure 3, but with the firewall enabled. **Figure 11** depicts how the firewall blocked the incoming packets that were previously allowed.

Figure 11: ICMP packets with the firewall

```
C:\WINDOWS\system32\cmd.exe
C:\Users\ygreall>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : nait.ca
    Link-local IPv6 Address . . . . . : fe80::38dd:dd99:6bca:3d7fa15
    IPv4 Address. . . . . : 10.131.12.244
    Subnet Mask . . . . . : 255.255.254.0
    Default Gateway . . . . . : 10.131.13.254

Ethernet adapter PLC:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::2570:8adb:743d:8e3c411
    IPv4 Address. . . . . : 172.16.50.76
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

C:\Users\ygreall>ping 172.16.50.250

Pinging 172.16.50.250 with 32 bytes of data:
Reply from 172.16.50.250: bytes=32 time=1ms TTL=63
Reply from 172.16.50.250: bytes=32 time=1ms TTL=63
Reply from 172.16.50.250: bytes=32 time=1ms TTL=63
Reply from 172.16.50.250: bytes=32 time=1ms TTL=63

Ping statistics for 172.16.50.250:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\Users\ygreall>
C:\Users\ygreall>ping 172.16.50.250

Pinging 172.16.50.250 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 172.16.50.250:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\ygreall>
```

Without Firewall

With Firewall

4.0 FIELD DEVICES

The next component of SCADA system is the simulated field equipment used in the lab for the simulation of the SCADA system. Field devices collect and capture the data from motors, sensors, and equipment in real time. Industries can use a variety of field devices and controllers depending on the application they can be implemented or removed from a network. The data that the field devices collect must be monitored and having the ability to see the data remotely is essential to keeping multimillion operations running. The projects plant devices utilized the Powerflex 70 and 753 VFDs and Control Logix PLCs as its equipment utilized for instructions within NAIT.

3.1 PowerFlex 70 EC-E Variable Frequency Drive (VFD)

The VFD is a field device that is used to monitor, control, and protect electric motors used in industries. It allows for precise speed control and torque regulation of processes, reducing the stress of startup and operation. The data monitored by the VFD gives a perspective into the performance and operational information of the electric motor/generator.

Figure 12:Power Flex 70



The Powerflex 70 VFD shown in Figure 10 is a pre-existing device within the lab. The Powerflex 70 features an optional single port ethernet NIC called 20-COMM-E shown in **Figure 13**. The 20-COMM_E module utilizes a DPI (Device port interface) which is a serial communication system that is proprietary to Rockwell automation.

Figure 13: 20-COMM-E



Communication on the network with the Powerflex 70 requires a change of certain parameters/settings of the NIC. Similarly to any device on a network, the Powerflex 70 requires an IP address and subnet mask to be identifiable within the network. The VFD parameter/settings for Ethernet is within the NIC in port 5 which is labeled as 20-COMM-E and is terminated to an ethernet cable. Configuration requires selecting the NIC in port 5 to access its networking parameters as shown in **Table 1**.

Table 1: 20-COMM-E settings

Parameter #	Setting	Description
Parameter #1	5	DPI port setting
Parameter #3	0	Disable BOOTP server
Parameter #4-7	IP	IP address
Parameter #8-11	Subnet	Subnet
Parameter #12-15	Gateway	Gateway

Note: Values from Allen Bradley 20-COMM-E EtherNet/IP Adapter User Manual Rockwell Automation. (2015)

In addition to setting the NIC, the DPI port controlling the devices must be changed and configured to allow communication and control from remote devices connected through the network and not by the HMI display on the VFD.

Table 2: Power Flex 70 Parameters

Parameter #	Setting	Description
Parameter #90	22 DPI Port 5	Speed reference selection
Parameter #93	22 DPI Port 5	Speed reference selection
Parameter #276	Bit 0,1 set to 0	Disable Port 1 and digital in logic mask
Parameter #276	Bit 5 Set To 1	Enabling port 5 logic mask
Parameter #277	Bit 5 Set To 1	Enabling port 5 start mask
Parameter #278	Bit 5 Set To 1	Enabling port 5 jog mask

Note: Values from Allen Bradley PowerFlex 70 Adjustable Frequency AC Drives User Manual Rockwell Automation. (2018)

Sending an ICMP packet to the IP address confirms if the connection is established between the Powerflex 70 and the computer. If this communication is established, the drive is ready to be added to the network. The Ethernet cable from the VFDs NIC (network interface card) is then able to be interfaced with a network. In the project the VFD is connected to the Mguard simulation of the plant network.

The Powerflex 70 data is then able to be sent through the network to the software Studio 5000 onto the PC. This is typically where the data is auto populated into data tags when the user defines the VFD in the software. Studio 5000, however had difficulty recognizing the VFD and only certain predefined data tags were able to be uploaded and recognized like motor start, stop, and speed reference input. Issues occurred with the other custom data tags called data links for the device where the user can select output voltage, output current, and output frequency tags to be shown. The diagnosis of these issues is covered in the addition Rockwell support section of this report.

3.2 Power Flex 753 Variable Frequency Drive (VFD)

Figure 14: PowerFlex 753



Industries use more than one field device on a network, emulating this the new prototype PowerFlex 753 (VFD) (**Figure 14**) that has not yet been into the lab was implemented into the project. The Powerflex 753 was implemented to simulate another device on the network and tested to alleviate the Powerflex 70s inability to define the custom datalink tags. Incorporation of the VFD required matching the motor nameplate specification with the VFD settings. To simplify and streamline the setup process the VFDs HMI has a quick setup menu that goes though the required parameter on PG. 17 of Allen Bradley PowerFlex 750-Series AC Drives Programing Manual (Rockwell Automation 2024, November). Following the set, the PowerFlex 753 VFDs NIC settings need to be configured. The Powerflex 753 utilizes an optional Dual port ethernet NIC called 20-750-ENETR shown in **Figure 15**. The 20-750-ENETR uses a PCIE connection to the motherboard of the drive.

Figure 15: 20-750-ENETR



The settings and all the network related parameters for the NIC are in the file labeled as Ethernet/IP (Port 13) on the HMI. The parameters that were changed are listed in **Table 3** which gives the parameter numbers, what the parameter should be set to and a brief description of what it is changing.

Table 3: Power Flex 753 Parameters

Parameter #	Setting	Description
Parameter #37	1	Parameter Set IP
Parameter #38-41	IP	IP address
Parameter #42-45	Subnet	Subnet
Parameter #46-49	Gateway	Gateway

Note: Values from Allen Bradley PowerFlex 750-Series AC Drives Programming Manual (2024, November). Rockwell Automation.

The NIC needs to be rebooted on the VFD to finalize changed settings. Sending an ICMP packet to the VFD from a PC confirms if a connection is established. The programming logic software Studio 5000 is then used to recognize the data tags from the VFD. Issues occurred with the module discovery where the drive wasn't listed in the catalog, this will be covered in the additional Rockwell support section.

4.0 PROGRAMING SOFTWARE CONFIGURATION AND SETUP

This section covers the software used to communicate with the A185 Lab prototype. Scada Systems can use a multitude of different software solutions for each device. The project utilizes Allen Bradley Rockwell equipment. Rockwell software suite was incorporated to allow for configuration of the Rockwell equipment over and on the network.

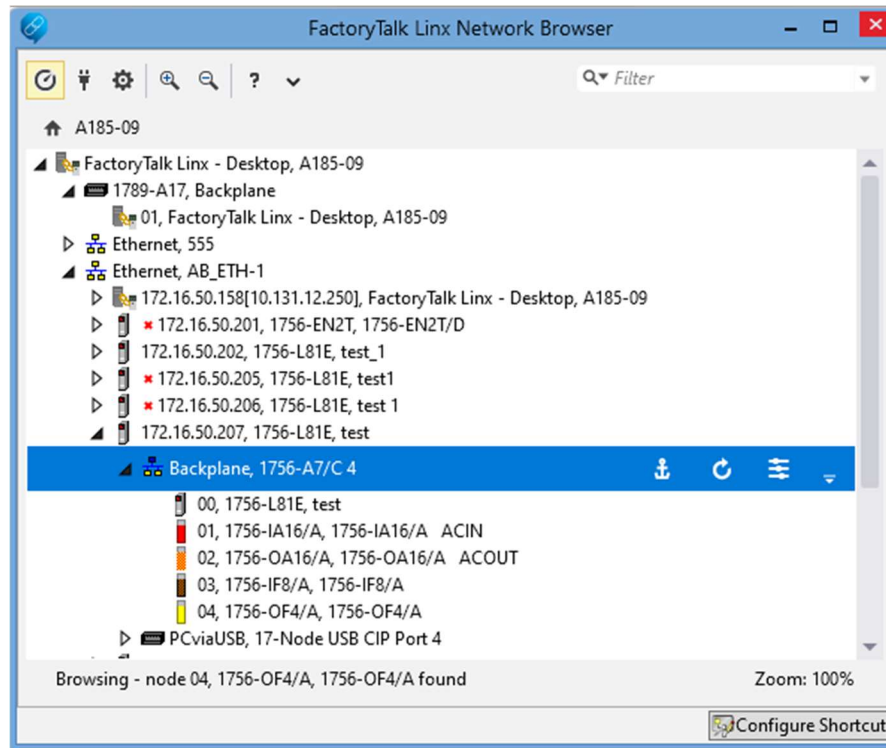
4.1 Rockwell Software suites

Rockwell software suites include RS links, studio 5000, Studio logics Designer are the main programs used to communicate with the Rockwell equipment. There are preexisting applications within the lab with a multitude of functionality.

4.1.1 Rockwell Network Browsers

RSlinx Classic and Factory Talk Linx is a communication software platform developed by Rockwell automation that enables communication paths and drivers to be created between PLCs, studio 5000 and other applications. Factory Talk Linx is the more modern version of RSlinx Classic and is used on non-legacy/older fields device. Both programs are considered network browsers that broadcast messages to find devices on the network. **Figure 16** shows the different devices that Factory Talk Linx can broadcast to on the local subnet. The network browser auto uploads electronic data sheets (EDS) when devices are added onto the network. The EDS files allow the computer to identify the information being sent though the network. This software was essential to establish and ensure that the communication path for the field devices was active. The network browsers were used to establish if the data from the field devices controlling the motor process is able to be defined by Allen Bradley software.

Figure 16: Factory Linx Network browser



4.1.2 STUDIO 5000

Studio 5000 is software developed by Rockwell Automation for programming/configuring Allen Bradley PLCs for industrial control systems. It uses tag-based architecture to store data, and parameter sets from field devices and allows for IEC 61131-3 Standard programming languages. This includes ladder logic function block diagram structure text and sequential flow chart. The prototype required us to use the software to create basic logic to monitor and control the VFDs. The PLC also stores real time information from a multitude of VFDs into tags which acts like a hub for operational data. This is also where we connect to from our SCADA host.

Once launched a connection must be established to the PLC to configure the NIC settings. This can be done via USB connection because to utilize the ethernet port the NIC parameters must be configured to be visible on the network.

The connection tab on the top of the page will drop down to see who's active and which network browser specified for seeing who's active shown in Figure 11. Opening who's active will launch the network browser specified and from there a drop down into the different modules visible on

the network will show similar to Figure 11. Once connected, the NIC IP Address, subnet and gateway PLC settings can be changed.

Removing the USB connection and sending ICMP packets confirms if the device is on the same network.

Once the connection is verified to be on the network, defining the VFD module is done by adding it via left clicking ethernet and clicking module discovery. Then search for the module in the catalog once to find and add the device. The VFD module prompt will display after putting in the VFD's ratings, revision and IP address. In the project, the Powerflex 753 and Powerflex 70 prompt is shown in **Figure 17** as well as **Figure 18** in Studio 5000.

Figure 17: Studio 5000 753 Connection page

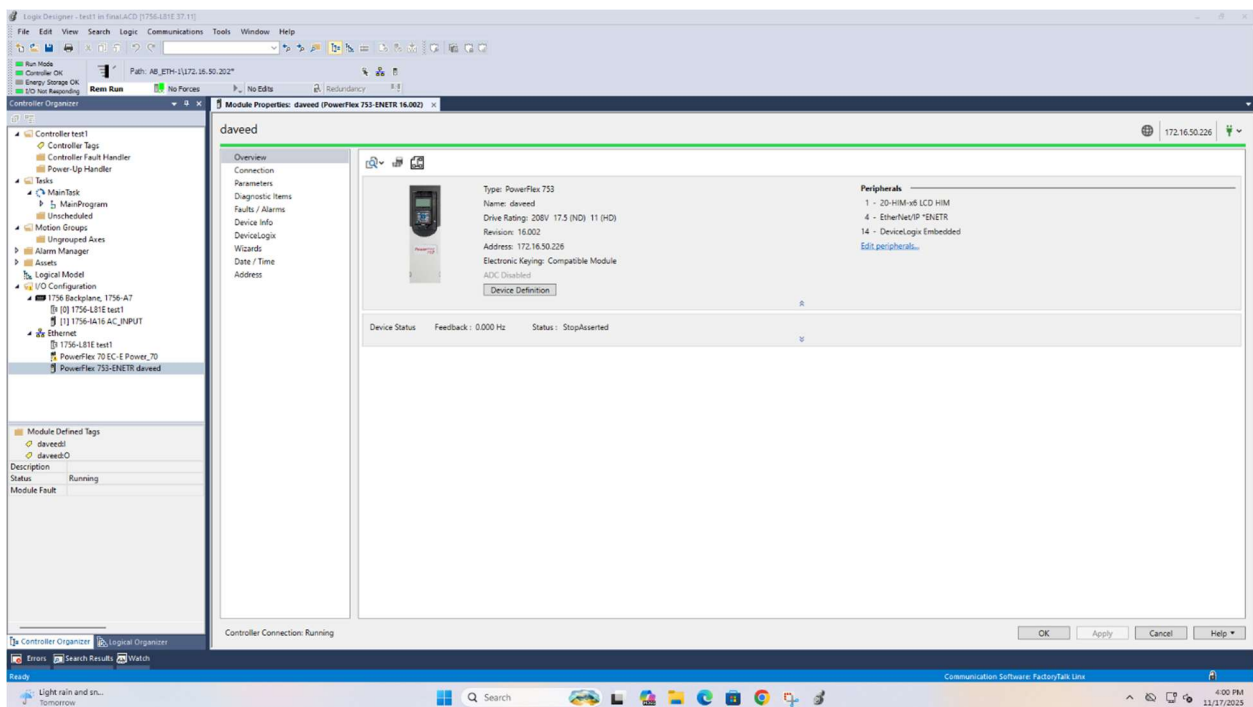
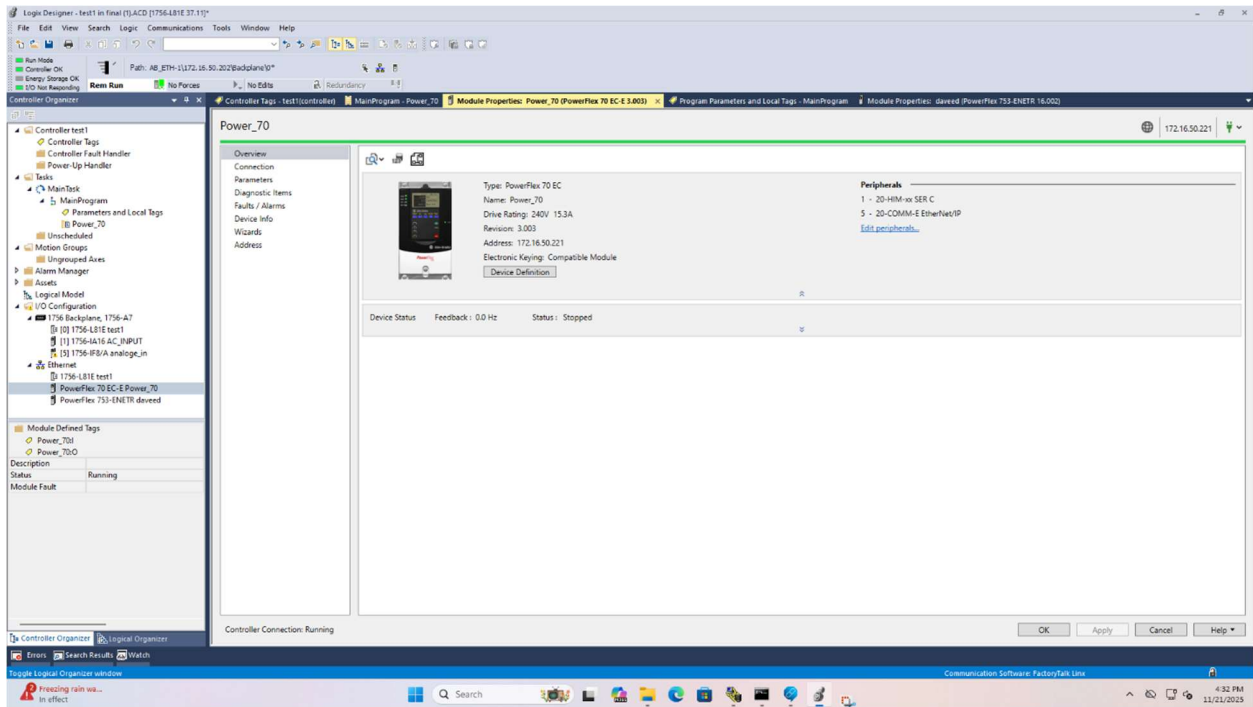


Figure 18 Studio 5000 70 Connection page



4.2 ADDITIONAL ROCKWELL SUPPORT

The addition of certain drivers, AOPS, EDS, and files may be required to operate added to the SCADA host PC to make Studio 5000 operation. This is the case with the project where issues occurred with the module discovery catalogue. The lab computers were missing these essential additions for the VFDs. This section specifies some of the support applications that may need to be installed to alleviate any issues.

4.2.1 Drive Tools

Drive Tools is a suite add on from Rockwell Automation used to configure, program, monitor and tune Allen Bradley VFDs. Drive Tools include software programs that allow programming, maintenance, and troubleshooting of power flex variable frequency drives. (Rockwell automation, 2025)

Drive executive is an application within drive tools that allows for an entire parameter set to be pulled off the drive onto a PC as shown in **Figure 19**. This is key as this application works with RXLink Classic to be able to have the parameters visible. This was used to test if the drive was communicating fully and attempt to have Studio 5000 recognize the VFD's. Through testing, Drive Tools didn't allow for Studio 5000 recognition however it was able to shows that the drive was able to communicate and send data across the network.

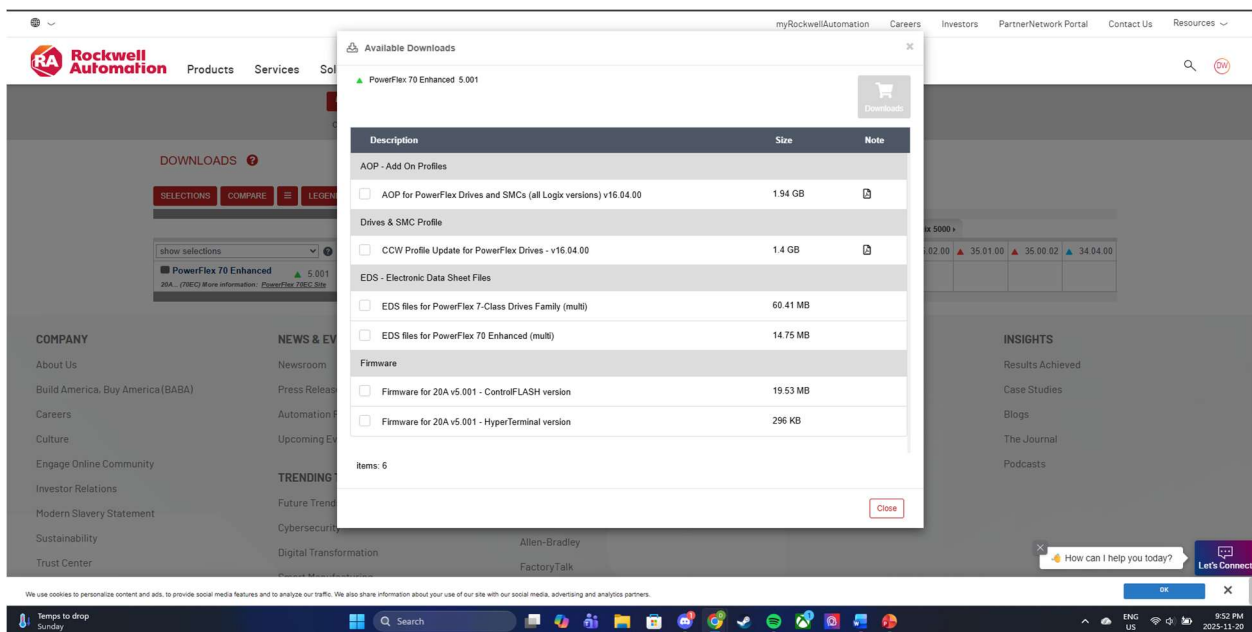
Figure 19: Drive Executive

#	Parameter Name	Value	Units	Internal Value	Comment	Default	Min	Max	Alias
1	Output Frequency	0.00	Hz	0x00000000		0.00	0.00	650.00	650.00
2	Commanded SpdRef	60.00		0x42700000		0.00	-11680.00	11680.00	
3	Mtr Vel Fdbk	0.00		0x00000000		0.00	-11680.00	11680.00	
4	Commanded Trq	0.00	%	0x00000000		0.00	-800.00	800.00	
5	Torque Cur Fdbk	0.00	Amps	0x00000000		0.00	-35.00	35.00	
6	Flux Cur Fdbk	0.00	Amps	0x00000000		0.00	-35.00	35.00	
7	Output Current	0.00	Amps	0x00000000		0.00	0.00	35.00	
8	Output Voltage	0.00	VAC	0x00000000		0.00	244.50	650.00	
9	Output Power	0.00	kW	0x00000000		0.00	0.00	3000.00	
11	DC Bus Volts	348.68	VDC	0x43A570A		0.00	0.00	460.00	
20	Rated Volts	208.00	VAC	0x43500000		0.00	0.00	690.00	
21	Rated Amps	17.50	Amps	0x418C0000		0.00	0.00	500.00	
22	Rated kW	4.00	kW	0x40800000		0.00	0.00	400.00	
25	Motor NP Volts	208.00	VAC	0x43500000		200.00	23.00	230.00	
26	Motor NP Amps	14.10	Amps	0x4161999A		14.20	0.10	175.00	
27	Motor NP Hertz	60.00	Hz	0x42700000		60.00	2.00	650.00	
28	Motor NP RPM	1760.0	RPM	0x44DC0000		1460.0	1.0	40000.0	
29	Mtr NP Pwr Units	HP		0		kW	HP		
30	Motor NP Power	5.00	kW	0x40A00000		4.00	0.01	2000.00	
31	Motor Poles	4	Pole	4		4	2	200	
33	Motor Ctl Mode	Induction SV		1		Induction SV	InductionVHz	IPM FV	
36	Maximum Voltage	230.00	VAC	0x43600000		200.00	50.00	230.00	
37	Maximum Freq	70.00	Hz	0x428C0000		130.00	0.01	650.00	
38	PWM Frequency	12.0	kHz	0x41400000		4.0	2.0	12.0	
65	VHz Curve	Custom V/Hz		0		Custom V/Hz	Custom V/Hz	Fan/Pump	
70	Autotune	Calculate		1		Calculate	Ready	Inertia Tune	
71	Autotune Torque	50.00	%	0x42400000		50.00	0.00	200.00	
150	Digital In Cfg	Run Edge		0		Run Edge	Run Edge	Run Level	
155	DI Enable	Disabled		0		19999.15	0.00	19999.15	
156	DI Clear Fault	Disabled		0		19999.15	0.00	19999.15	
157	DI Aux Fault	Disabled		0		19999.15	0.00	19999.15	
158	DI Stop	Disabled		0		19999.15	0.00	19999.15	
159	DI Cur Limit Stop	Disabled		0		19999.15	0.00	19999.15	
160	DI Coast Stop	Disabled		0		19999.15	0.00	19999.15	
161	DI Start	Disabled		0		19999.15	0.00	19999.15	
162	DI Fwd Reverse	Disabled		0		19999.15	0.00	19999.15	
163	DI Run	Disabled		0		19999.15	0.00	19999.15	
164	DI Run Forward	Disabled		0		19999.15	0.00	19999.15	
165	DI Run Reverse	Disabled		0		19999.15	0.00	19999.15	
166	DI Jog 1	Disabled		0		19999.15	0.00	19999.15	
167	DI Jog 1 Forward	Disabled		0		19999.15	0.00	19999.15	
168	DI Jog 1 Reverse	Disabled		0		19999.15	0.00	19999.15	
169	DI Jog 2	Disabled		0		19999.15	0.00	19999.15	
170	DI Jog 2 Forward	Disabled		0		19999.15	0.00	19999.15	
171	DI Jog 2 Reverse	Disabled		0		19999.15	0.00	19999.15	
172	DI Manual Ctl	Disabled		0		19999.15	0.00	19999.15	

4.2.2 AOP's, EDS, And Firmware

Add on profiles (AOPs) and electronic data sheets (EDS) files are not a specific software tool but are additions that allow for communication to modules and are critical for integration drives into PLCs. The AOP contains information and configurations about the device so the PLC can communicate with it. EDS file for a particular device contains information about that device and is required for a PLC to communicate with that device (Shawn Dietrich, 2023). This can all be accessed by looking at Rockwell Automations setup that also includes Connected Components Workbench CCW profile updated drives for Powerflex drives, and firmware as shown in **Figure 20**. The lab computers were missing the AOPs which caused communication issues with the VFDs. The installation of the AOPs by the IT department at NAIT allowed the VFDs to communicate a live parameter set across the network.

Figure 20: Rockwell Automation Files For VFD



Rockwell automation Downloads page (2025).

Once the files are downloaded, they will have to be opened and executed to be added to the SCADA Host.

4.2 Ignition

Ignition is modern software platform specifically designed for monitoring/ controlling industrial automation SCADA systems (factory, utilities, building automation). With the ability of web-based deployment. The program operates with a range of platforms including Windows, MacOS, and Linux OS. The Program has Ignition Designer where individuals can fully design an GUI to a plants requirements and includes Ignition Perspective to have the GUI displayed. Ignition also includes features for data collection/logging with its ability to connect to structured query language SQL databases. These are massive data storage devices that record the history of devices such as tags logs alarms, and trend data. The project utilizes Ignition Designer and Ignition Perspective to be able to simulate a SCADA host that would be monitoring and controlling the plant floor across a network.

Firstly, the Logix controller PLC is added onto the LAN side of the network and the SCADA host with Ignition on an adjacent network. Connection is verified by sending an ICMP packet to the device to see if the SCADA host can reach the Logix controller PLC. Once ignition is launched access to the gateway will appear once logged in. The addition of the PLC as a device can be done under the connections tab. The IP address of the PLC can then be entered into the menu as shown in **Figure 21** below.

Figure 21: Ignition Adding Devices

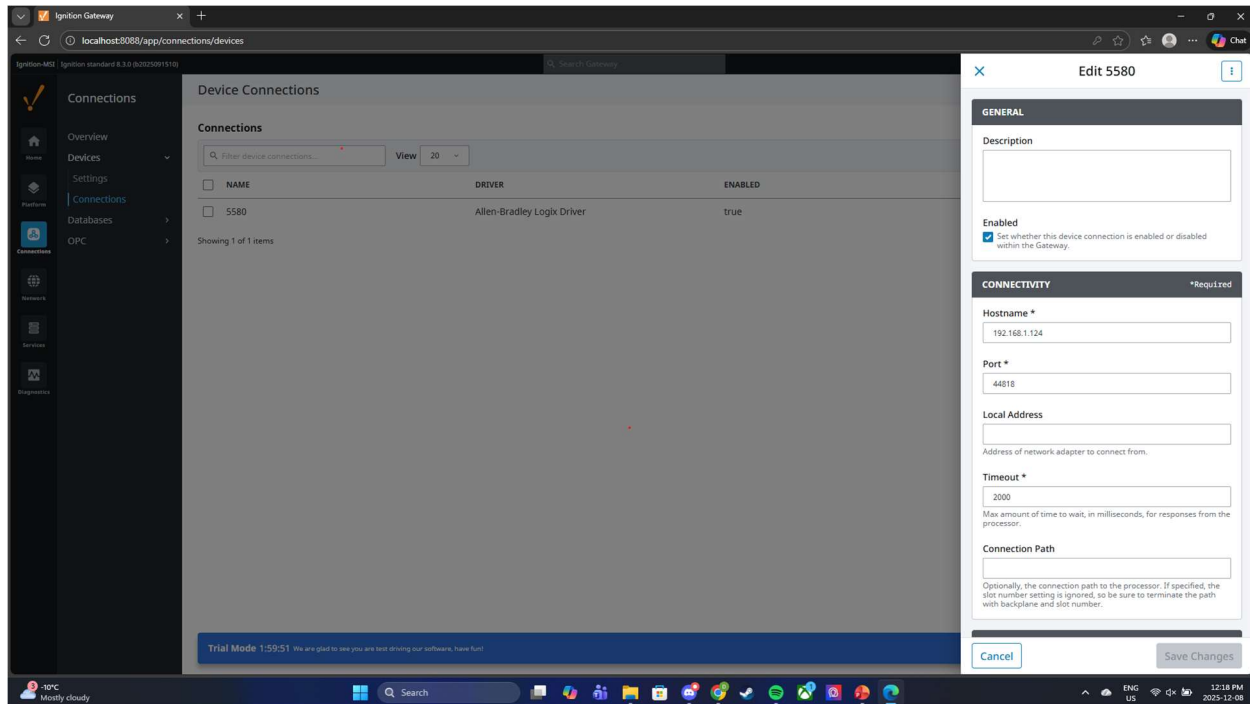


Figure 22: Ignition Device Connected



Once the green check appears the device is added and connected onto the network as shown in **Figure 22**. Ignition designer can then be launched to begin an GUI design.

NOTE: The prototype GUI designed is very simple and does not follow any specific design IEC or ISA standards for GUI however this can be implemented per the specification of the application.

In side of the designer shown in Ignition Designer, begin by seeing if tags are visible by using the tag browser on the left **Figure 23**. If not, then tags will have to be added by pressing the plus arrow and opening the dropdowns for the PLC tags shown in **Figure 24** below. These tags from the PLC can be added to Ignition Designer. These tag values hold the live data that represents the motors

status on the adjacent network. Once a design is complete it can be saved and opened on Ignition Perspective. Ignition Perspective allows for the design to be launched as a GUI and viewed.

Figure 23: Ignition Perspective

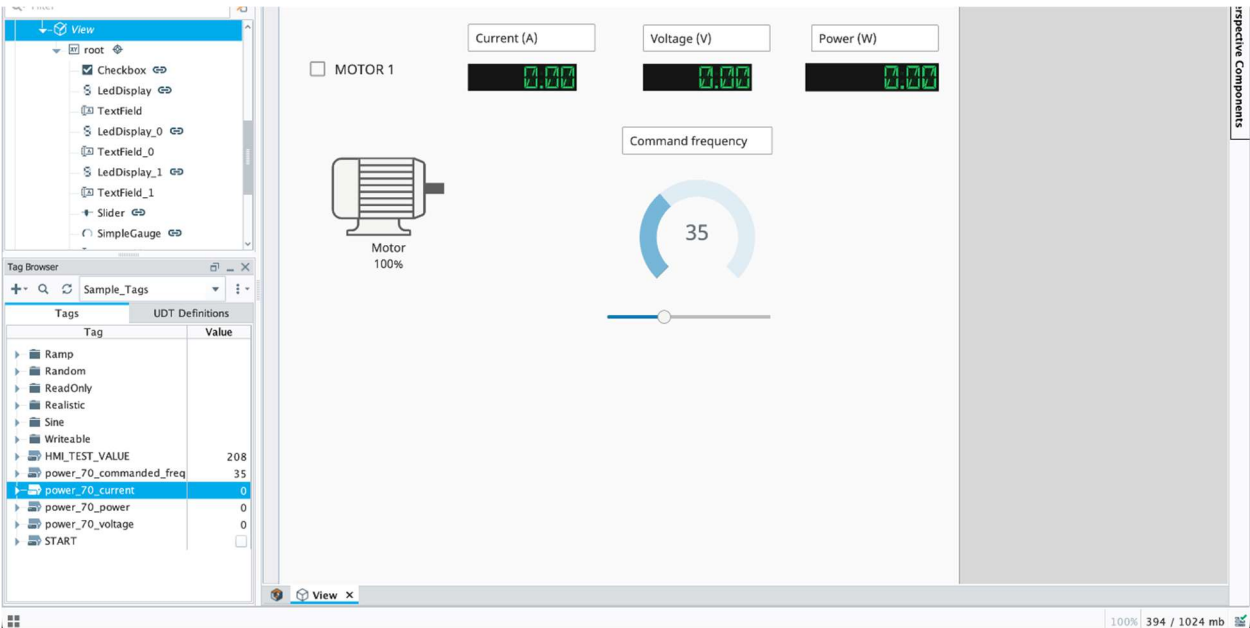
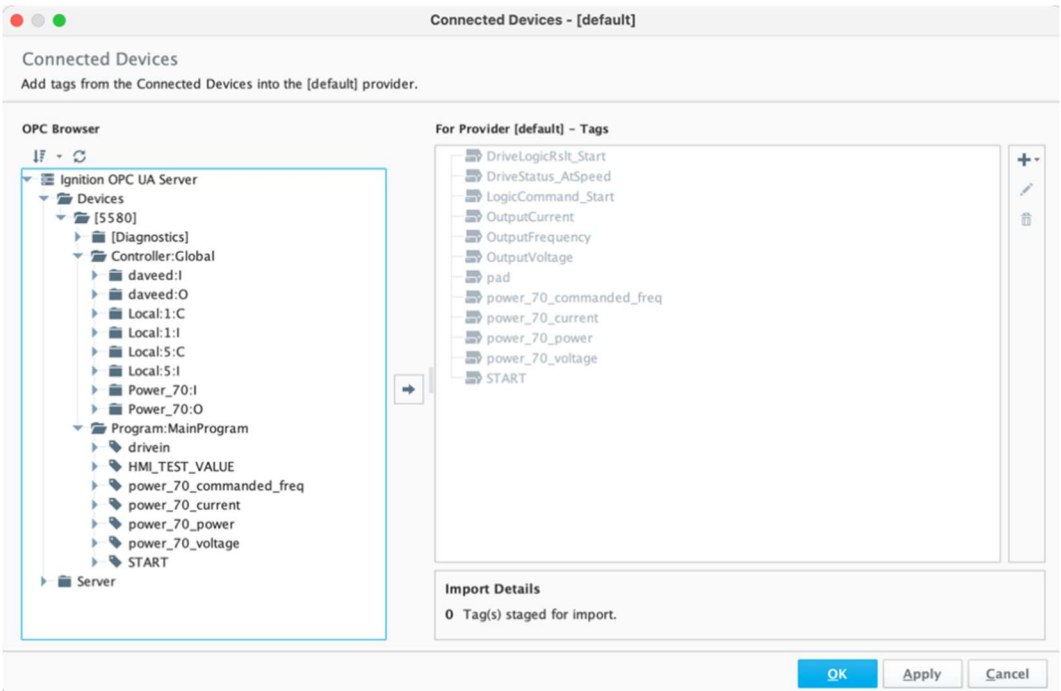


Figure 24: Tag Brower



5.0 CONCLUSION

The result of this demonstration confirms that a fully operating SCADA model in lab A185 is achievable with the ability to establish commutation among all the equipment across a network. Specifically demonstrating data being pulled from multiple VFDs into the PLC to then be sent across a network to be controlled remotely.

Through effective research and testing with router equipment the network was able to establish an effective network. The simulation shows 1:1 NAT being done between the PLC in LAN and the computers in the control network on WAN using virtual IP addresses. In addition, is a working static route between the SCADA network and the Floor network. This successfully achieved a complete network that can send and receive data.

Additionally, with effective troubleshooting the data link tag we get a better understanding of Rockwell Automations Studio5000 and the setup of multiple field devices. This was accomplished with the addition of AOP and EDS files executed on the computer from the Rockwell website.

Lastly is the successful creation of a GUI on Ignition. Though extensive trial and error the project was able to have a simple yet functional GUI that demonstrates data from field devices being sent across the network to a remote host.

In the end, the demonstration was achieved the preset goals with data transferring through multiple networks and with Ignition an GUI was made that could present the gathered data in real time. Additions such as adding in OPC server, and a VPN tunnel could be added for data acquisition and remote access to help simulate a more real-world plant network can be considered. Networks are expandable by application and the prototype achieved demonstrates a successful replication of an Industrial ethernet network.

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Appendix A - Definitions

AOP: Add On Profiles is a device specific configuration package used by studio 5000 to allow for integration of field devices.

EDS: Electronic Data Sheets is a data sheet that specifies how to communicate with a device and interacts with configuration software

GUI: Graphical User Interface (GUI) an interface that connects a physical user to a device allowing users to monitor and control machinery remotely through web-based deployment.

HMI: Human Machine Interface (HMI) an interface that connects a physical user to a device allowing users to monitor and control machinery.

Host: Any device that is connected to a network that can communicate to other devices.

ISP: Internet Service Provider (ISP) is a company that provides individuals and organizations with access to internet.

LAN: Local Area Network (LAN) that connects devices in a small/limited area to share data with each other.

NIC: Network interface card (NIC) is a hardware interface for devices allowing data to be sent or received via wired or wireless connection.

Ping: It is a tool that tests the reachability of a destination host within the network by sending ICMP packets

PLC: Programmable logic controllers are industrial computers that automate and control multiple processes by using user defined logic.

Private IP: A unique identifier number assigned to every device within a LAN that is not visible from the internet.

Public IP: A unique identifier provided by an ISP to a router allowing hosts in the LAN to communicate with the internet.

WAN: Wide Area Network (WAN) is a network that connects multiple small networks (LAN) over a large geographical area.